

Practical Work 2: RPC File Transfer

Nguyen Trong Bach
Student ID: 23BI14057

December 5, 2025

1 Introduction

This practical upgrades the TCP file transfer system from Practical Work 1 into an RPC-based system. Remote Procedure Call (RPC) allows a client to call functions on the server as if they were local. Python's XML-RPC module is used due to its simplicity and built-in support.

2 RPC Service Design

The RPC server exposes four methods:

- `list_files()` — lists available files
- `upload_file(filename, data)` — uploads a file
- `download_file(filename)` — returns file contents
- `delete_file(filename)` — removes a file

The architecture is shown in Figure 1.

3 System Organization

The system is organized into:

1. **server.py**: RPC server implementing all file operations.
2. **client.py**: client application calling RPC functions.
3. **server_storage/**: server-side directory storing files.

4 Implementation

Below is an example from the server:

```
def upload_file(filename, data):  
    filepath = os.path.join(BASE_DIR, filename)  
    with open(filepath, "wb") as f:  
        f.write(data.data)  
    return True
```

5 Group Work

This practical was completed individually.

6 Conclusion

RPC simplifies networking by allowing the client to use remote functions as if they were local. The implementation successfully upgrades the TCP-based solution to an RPC-based file transfer system.